Every odd-indexed term of OEIS A389472 after the first is even

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Abstract

OEIS A389472 is defined by the functional equation $A(x) = A(x^2 + x^3)/x^2 - 1$ for its ordinary generating function, and carries a conjecture of P. D. Hanna that $a(2n-1)$ is even for every $n > 1$. It is true. The equation determines the coefficients one at a time, so it determines them modulo 2; and over $\mathbb{F}_2$ it turns out to be *parity-preserving*: the series $x + C(x^2)$ satisfies it exactly, for C the solution of the companion equation $y\,C(y) = C(y^2 + y^3)$, because $(x^2 + x^3)^2 = x^4 + x^6$ in characteristic 2. Uniqueness then forces the reduction of A to be supported, apart from the term x, on even exponents only — which is the conjecture. The entry's second conjecture, modulo 3, is *not* settled here; Section 5 says where the argument fails to carry over.

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1 The sequence and the conjecture

OEIS A389472 is "G.f. satisfies A(x) = A(x^2 + x^3)/x^2 - 1.". It has offset 1 and begins

$$1, 1, 2, 3, 6, 9, 14, 24, 42, 72, \ \ldots$$

The entry gives the expansion

G.f.: A(x) = x + x^2 + 2*x^3 + 3*x^4 + 6*x^5 + 9*x^6 + 14*x^7 + 24*x^8 + ...

so, writing $A(x) = \sum_{n\geq 1} a(n)x^n$, the entry fixes

$$a(1) = 1, \qquad a(2) = 1, \tag{1}$$

and defines the remaining coefficients by $A(x) = A(x^2+x^3)/x^2-1$, which we clear of the division and write as

$$x^2\big(A(x) + 1\big) \ = \ A\big(x^2 + x^3\big). \tag{2}$$

This is the only input taken from the entry.

The entry separately carries the following comment:

Conjecture: a(2*n-1) = 0 (mod 2) for n > 1. — *Paul D. Hanna*

As of the "Last modified" line on the live entry (Oct 17 2025 00:08:59, revision 7) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way. The entry carries a second conjecture of the same contributor, that $a(3n-1) \equiv 0$ modulo 3 for $n > 1$. It is not proved below and it is not claimed.

2 The equation determines the sequence

Because $(x^2+x^3)^n = x^{2n}(1+x)^n$, comparing coefficients of x^N in (2) gives, for every $N \geq 2$,

$$a(N-2) \;=\; \sum_{n\geq 1} a(n)\binom{n}{N-2n}, \tag{3}$$

with $a(0) := 0$ and the convention $\binom{n}{j} = 0$ unless $0 \leq j \leq n$.

Lemma 1. *The coefficient $a(1) = 1$ is forced by (2). Given also $a(2)$, equation (3) determines $a(m)$ for every $m \geq 3$ as a polynomial with integer coefficients in $a(1), \ldots, a(m-1)$; so (1) and (2) together have exactly one integer solution.*

Proof. At $N = 2$, (3) reads $a(0) = a(1)\binom{1}{0}$ — reading the left side of (2) instead, the coefficient of x^2 there is 1, so $a(1) = 1$.

Now take $N \geq 5$ and put $m = N-2 \geq 3$. On the right of (3) the binomial $\binom{n}{N-2n}$ vanishes unless $2n \leq N$, so only $a(n)$ with $n \leq \lfloor N/2 \rfloor$ occur. Since $N \geq 5$,

$$\left\lfloor \tfrac{N}{2} \right\rfloor \;\leq\; N-3 \;=\; m-1,$$

so the right side involves only $a(1), \ldots, a(m-1)$ and (3) expresses $a(m)$ through them. (At $N = 3$ and $N = 4$ the relation reads $a(1) = a(1)$ and $a(2) = a(2)$, giving no information; that is why $a(2)$ is part of the data.) Induction finishes the proof. □

Corollary 2. *Let $k \geq 2$. If $\bar{A} \in (\mathbb{Z}/k\mathbb{Z})[[x]]$ has $[x^1]\bar{A} = [x^2]\bar{A} = 1$, no constant term, and satisfies (2) with all coefficients read modulo k, then $a(n) \equiv [x^n]\bar{A} \pmod{k}$ for every n.*

Proof. Reduction modulo k is a ring homomorphism, so it carries (3) to the same relation over $\mathbb{Z}/k\mathbb{Z}$; by Lemma 1 that relation, together with the two initial values, has a unique solution, and both $a \bmod k$ and the coefficients of $\bar{A}$ solve it. □

3 The companion equation

Work in $\mathbb{F}_2[[y]]$.

Lemma 3. *There is exactly one $C(y) = \sum_{m\geq 1} c_m y^m \in \mathbb{F}_2[[y]]$ with $c_1 = 1$ and*

$$y\,C(y) \;=\; C\big(y^2+y^3\big). \tag{4}$$

Proof. Comparing coefficients of y^m in (4) and using $(y^2+y^3)^n = y^{2n}(1+y)^n$ gives $c_{m-1} = \sum_{n\geq 1} c_n\binom{n}{m-2n}$, where again only $n \leq \lfloor m/2 \rfloor$ occur. For $m \geq 3$ we have $\lfloor m/2 \rfloor \leq m-2$, so this expresses c_{m-1} through $c_1, \ldots, c_{m-2}$; at $m = 2$ it reads $c_1 = c_1$. Hence c_1 may be prescribed and everything after it is determined. □

Lemma 4. *Let C be as in Lemma 3 and put $\bar{A}(x) = x + C(x^2)$. Then $\bar{A}$ satisfies (2) over $\mathbb{F}_2$, and $[x^1]\bar{A} = [x^2]\bar{A} = 1$.*

Proof. In characteristic 2 squaring is additive, so

$$\big(x^2+x^3\big)^2 \;=\; x^4+x^6. \tag{5}$$

Substituting $u = x^2+x^3$ into $\bar{A}(u) = u + C(u^2)$ and using (5),

$$\bar{A}\big(x^2+x^3\big) \;=\; x^2+x^3+C\big(x^4+x^6\big),$$

while

$$x^2\big(\bar{A}(x)+1\big) \;=\; x^2\big(x+C(x^2)+1\big) \;=\; x^2+x^3+x^2C\big(x^2\big).$$

The two agree exactly when $x^2C(x^2) = C(x^4+x^6)$, which is (4) read at $y = x^2$. Finally $[x^1]\bar{A} = 1$ and $[x^2]\bar{A} = c_1 = 1$. □

4 The reduction modulo 2

Theorem 5. *$a(2n-1) \equiv 0 \pmod 2$ for every $n > 1$. This is the conjecture of Section 1.*

Proof. By Lemma 4 the series $\bar{A}(x) = x + C(x^2)$ satisfies (2) over $\mathbb{F}_2$ and matches (1). By Corollary 2 with $k = 2$ it is therefore the reduction of A modulo 2.

Every exponent occurring in $C(x^2)$ is even, so the only odd exponent occurring in $\bar{A}$ is 1. Hence $[x^m]\bar{A} = 0$ for every odd $m \geq 3$, that is, $a(m) \equiv 0 \pmod 2$ for every odd $m \geq 3$. Writing $m = 2n - 1$ with $n > 1$ gives the statement. $\square$

Remark 6. The mechanism is worth isolating: (2) substitutes $x^2 + x^3$, and in characteristic 2 the square of that is again a polynomial in x^2, by (5). So the substitution cannot manufacture new odd exponents beyond the ones already present, and the single odd exponent 1 forced by $a(1) = 1$ is all there is.

5 Where the argument stops

The entry's other conjecture is that $a(3n-1) \equiv 0$ modulo 3 for $n > 1$, that is, that the reduction of A modulo 3 is supported off the exponents $\equiv 2 \pmod 3$ apart from x^2 itself. That is not proved here, and the proof above does not adapt: it turns on the Frobenius identity (5), which in characteristic 3 becomes $(x^2+x^3)^3 = x^6+x^9$ — but (2) substitutes the *first* power of x^2+x^3, not the third, so the Frobenius map is not available where it would be needed. The residues modulo 3 were computed for every $n \leq 60$ and the conjecture holds throughout that range; no proof is offered.

6 Verification

Three checks, each independent of the algebra above.

First, the recursion (3) was run in exact integer arithmetic from (1) and compared against the entry's DATA at every one of the 41 published indices; the values agree exactly. A recursion that reproduced only some of them would mean the functional equation had been transcribed wrongly.

Second, the series C of Lemma 3 was generated from its own recursion over $\mathbb{F}_2$ and $x+C(x^2)$ compared, coefficient by coefficient, against $a(n)$ mod 2 for every $n \leq 60$; the two agree at every index, with no mismatch. Equation (4) was checked directly over $\mathbb{F}_2$ to order 60.

Third, $a(2n-1)$ mod 2 was evaluated for every n with $2n-1 \leq 60$: all zero, as the theorem requires.

References

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